Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	06 July 2026
Team ID	
Project Name	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

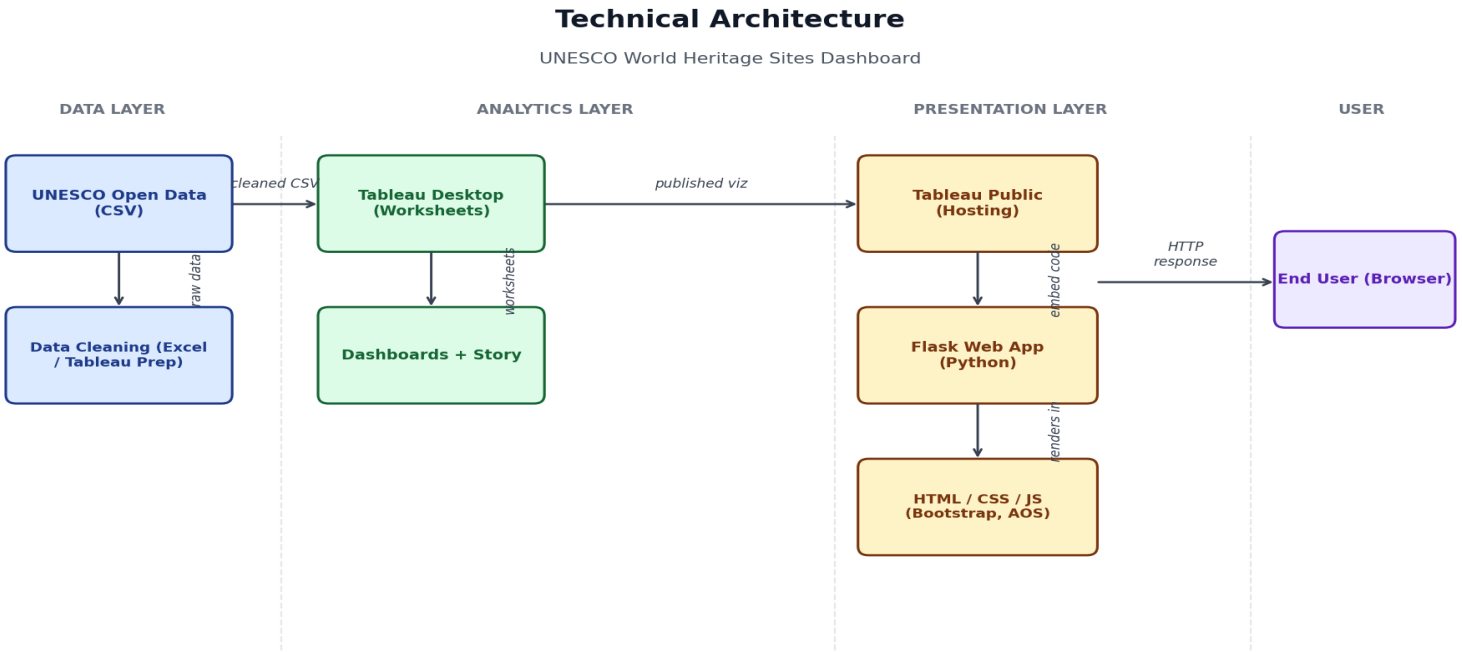


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How the user interacts with the dashboard and website	Tableau Dashboard UI, HTML, CSS, Bootstrap, JavaScript
2.	Application Logic-1	Cleans and prepares raw UNESCO site data for analysis	Excel / Tableau Data Prep
3.	Application Logic-2	Builds all chart worksheets and dashboards from the cleaned data	Tableau Desktop
4.	Application Logic-3	Serves the published dashboards inside a custom web page	Flask (Python)
5.	Database	Stores the cleaned heritage sites dataset used by all worksheets	Local data extract (.hyper) / CSV
6.	Cloud Database	Hosts the published interactive workbook for public access	Tableau Public Cloud
7.	File Storage	Stores the dataset, images and generated Word/PDF documentation files	Local Filesystem
8.	External API-1	Provides base map tiles for the geographic visualizations	Mapbox / OpenStreetMap (via Tableau Maps)
9.	External API-2	Supplies web page styling and animation libraries	Bootstrap CDN, Animate.css, AOS.js
10.	Machine Learning Model	Not applicable — the dashboard uses Tableau's built-in forecasting, not a custom ML model	Tableau built-in Forecasting (Exponential Smoothing)
11.	Infrastructure (Server / Cloud)	Local Server Configuration: Flask dev server on localhost, port 1212. Cloud Server Configuration: Tableau Public cloud hosting for dashboards/story	Local machine + Tableau Public Cloud

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used to style and animate the web front-end	Bootstrap 5, Animate.css, AOS (Animate on Scroll)
2.	Security Implementations	Published dashboards are read-only for public viewers; no personal or sensitive data is exposed since the dataset is public UNESCO information	Tableau Public read-only viz embedding
3.	Scalable Architecture	Simple layered architecture (data -> analytics -> presentation) allows new charts, years or regions to be added without redesigning existing dashboards	Layered/3-tier architecture (Data, Analytics, Presentation)
4.	Availability	Dashboards are hosted on Tableau Public's cloud infrastructure, giving high uptime independent of the local Flask server	Tableau Public Cloud Hosting
5.	Performance	Pre-aggregated worksheets and Tableau's in-memory data engine keep chart rendering and filtering fast even with 1,100+ site records	Tableau Hyper Engine, browser caching via CDN-hosted libraries